

Wireless Regional Area Network (WRAN) Using Cognitive Radio

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CERTIFICATE

Date: 28 April 2026

This is to certify that the Main Project work entitled “**Wireless Regional Area Network (WRAN) Using Cognitive Radio**” is a bonafide work carried out by

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in partial fulfillment of the requirements for the degree of **BACHELOR OF TECHNOLOGY** in **ELECTRONICS & COMMUNICATION ENGINEERING** by the Jawaharlal Nehru Technological University, Hyderabad during the academic year 2025-26.

The results embodied in this report have not been submitted to any other University or Institution for the award of any degree or diploma.

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ABSTRACT

The rapid expansion of wireless communication services has resulted in spectrum scarcity, particularly in rural and underserved regions where existing broadband solutions such as WiMAX face limitations in coverage, power requirements, and line-of-sight constraints. To address this issue, the IEEE 802.22 Wireless Regional Area Network (WRAN) standard was introduced as the first global standard to incorporate Cognitive Radio (CR) techniques, enabling opportunistic use of underutilized TV white spaces (54–862 MHz) for broadband wireless access. Earlier survey-based studies on WRAN have primarily focused on providing a conceptual overview, comparing its capabilities with WiMAX, and identifying critical challenges such as self-coexistence among operators and the hidden incumbent problem. These works established the theoretical foundation of WRAN technology and its potential applications in bridging the rural digital divide.

In this project, we advance beyond survey and theory by exploring practical implementation and performance evaluation of CR-based WRAN systems. Using MATLAB simulation, we investigate spectrum hole detection, efficient frequency allocation, dynamic spectrum access, and user assignment strategies to optimize spectrum utilization. Furthermore, hardware validation is carried out using USRP-based setups, demonstrating the adaptability of CR systems in real-time conditions. The project also addresses security aspects of IEEE 802.22 networks, highlighting potential vulnerabilities and proposing approaches to enhance robustness in dynamic spectrum environments.

Thus, while previous work offered a broad survey and comparative analysis of WRAN technology, our project contributes by providing simulation-driven, prototype-oriented insights that move closer to real-world deployment of broadband solutions for rural connectivity.

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ABBREVIATIONS

CR	– Cognitive Radio
WRAN	– Wireless Regional Area Network
DSA	– Dynamic Spectrum Access
PU	– Primary User
SU	– Secondary User
SNR	– Signal-to-Noise Ratio
AWGN	– Additive White Gaussian Noise
FFT	– Fast Fourier Transform
PSD	– Power Spectral Density
CFAR	– Constant False Alarm Rate
ROC	– Receiver Operating Characteristic
Pd	– Probability of Detection
Pfa	– Probability of False Alarm
IEEE	– Institute of Electrical and Electronics Engineers
CPE	– Customer Premises Equipment
IoT	– Internet of Things
NLOS	– Non-Line-of-Sight
μ (Mu)	– Mean
σ (Sigma)	– Standard Deviation

CHAPTER 1: INTRODUCTION

CHAPTER 1: INTRODUCTION

1.1 Introduction

The rapid and exponential growth of wireless communication services has significantly increased the demand for radio frequency spectrum. With the widespread adoption of mobile devices, Internet of Things (IoT) applications, smart communication systems, and high-speed broadband services, the need for efficient spectrum utilization has become more critical than ever. However, traditional spectrum allocation policies follow a fixed and static approach, where frequency bands are licensed to specific users or services for exclusive use. This rigid allocation leads to inefficient utilization of the spectrum, as several licensed bands remain underutilized across different time periods and geographical locations, while other bands experience heavy congestion.

Studies conducted by regulatory authorities have shown that a large portion of the allocated spectrum is not used continuously, leading to the existence of unused frequency bands known as “spectrum holes” or “white spaces.” This mismatch between spectrum scarcity and underutilization has motivated researchers to develop intelligent and adaptive spectrum management techniques that can dynamically utilize available resources without causing interference to licensed users.

Cognitive Radio (CR) is an advanced wireless communication paradigm that addresses this challenge by enabling dynamic and efficient spectrum utilization. A Cognitive Radio system is capable of sensing its surrounding radio environment, detecting unused spectrum bands, and adapting its transmission parameters such as frequency, power, bandwidth, and modulation schemes. This adaptability allows unlicensed users, known as Secondary Users (SUs), to opportunistically access the spectrum without causing harmful interference to licensed Primary Users (PUs). The key functionalities of a Cognitive Radio system include spectrum sensing, spectrum decision, spectrum sharing, and spectrum mobility, all of which contribute to intelligent and efficient spectrum usage.

One of the most important features of Cognitive Radio is its ability to perform real-time spectrum sensing. This involves continuously monitoring the radio environment to detect the presence or absence of Primary Users. Based on this information, the system identifies available spectrum opportunities and dynamically adjusts its transmission strategy. This real-time adaptability makes Cognitive Radio highly suitable for modern wireless networks, where spectrum usage patterns are highly dynamic and unpredictable.

In addition to improving spectrum efficiency, Cognitive Radio plays a crucial role in enabling heterogeneous wireless communication systems. With the coexistence of multiple technologies such as cellular networks, Wi-Fi, and IoT devices, efficient spectrum sharing becomes essential to avoid interference and ensure reliable

communication. Cognitive Radio facilitates seamless coexistence by intelligently allocating spectrum resources based on real-time conditions, thereby enhancing overall network performance.

A significant application of Cognitive Radio technology is in the IEEE 802.22 Wireless Regional Area Network (WRAN), which is the first global standard designed to operate using Cognitive Radio principles. IEEE 802.22 utilizes TV white space frequency bands ranging from 54 MHz to 862 MHz, which are typically underutilized by television broadcasting services. These frequency bands possess excellent propagation characteristics, including long-range coverage, low path loss, and strong penetration through obstacles such as buildings and vegetation. As a result, WRAN systems are highly suitable for providing broadband connectivity in rural and remote areas where traditional communication infrastructure is limited or unavailable.

The deployment of WRAN systems offers a promising solution to bridge the digital divide by delivering affordable and reliable internet access to underserved populations. By leveraging Cognitive Radio technology, WRAN can efficiently utilize available spectrum resources while ensuring protection of licensed users. This makes it an effective approach for extending connectivity to areas where conventional wired or wireless solutions are not feasible.

Despite its advantages, Cognitive Radio technology also faces several challenges. Accurate spectrum sensing is critical to avoid interference with Primary Users, and performance can be affected by factors such as noise uncertainty, multipath fading, shadowing, and the hidden node problem. Additionally, designing efficient algorithms that can operate in real-time with low computational complexity remains a key research challenge. Therefore, advanced techniques such as adaptive thresholding and multi-band sensing are required to enhance detection reliability and system performance.

In this project, the focus is on developing an efficient spectrum sensing mechanism for Cognitive Radio-based WRAN systems using multi-band energy detection combined with adaptive thresholding techniques. The proposed approach aims to improve detection accuracy under varying noise and channel conditions while maintaining low complexity. By enabling reliable identification of spectrum holes and efficient utilization of available bands, the system contributes to the advancement of Cognitive Radio technology and its practical implementation in next-generation wireless communication systems.

1.2 Aim of the Project

The main objective of this project is to design and simulate a Cognitive Radio-based spectrum sensing and access mechanism for WRAN systems. The proposed system aims to efficiently detect spectrum holes using multi-band energy detection combined with an adaptive thresholding technique based on Constant False Alarm

Rate (CFAR). Additionally, the system enables dynamic spectrum access for Secondary Users (SUs) while ensuring protection of Primary Users by avoiding interference. The goal is to achieve reliable detection, efficient spectrum utilization, and practical feasibility for real-world deployment.

1.3 Methodology

To achieve the objectives of the project, a MATLAB-based simulation model is developed that incorporates the key functional components of a Cognitive Radio system. The overall methodology focuses on spectrum sensing, adaptive thresholding, and intelligent decision-making for dynamic spectrum access.

The activity of the Primary User (PU) is modeled using a two-state Markov chain to simulate realistic ON/OFF behavior over time. This model captures the temporal variations in spectrum occupancy and provides a practical framework for evaluating sensing performance. The received signal, affected by channel conditions such as fading and noise, is analyzed using the Fast Fourier Transform (FFT) to enable multi-band spectrum sensing. The frequency spectrum is divided into multiple bands, and the energy in each band is computed to detect spectrum availability.

An adaptive thresholding mechanism based on the Constant False Alarm Rate (CFAR) technique is implemented to improve detection accuracy under varying noise conditions. The threshold is dynamically updated using statistical properties of noise samples collected over time, thereby maintaining a consistent false alarm rate. In addition, a distance-based interference avoidance mechanism is incorporated to ensure that Secondary User (SU) transmissions do not interfere with Primary Users.

Finally, the performance of the system is evaluated using key metrics such as energy variation, adaptive threshold behavior, band-wise activity of Primary and Secondary Users, and Receiver Operating Characteristic (ROC) curve analysis. These evaluations help in assessing the effectiveness and reliability of the proposed approach.

1.4 Significance of the Work

The proposed work presents an efficient and practical approach to spectrum sensing in Cognitive Radio systems, especially under challenging conditions such as low Signal-to-Noise Ratio (SNR) and noise uncertainty. By integrating multi-band energy detection with an adaptive thresholding technique based on Constant False Alarm Rate (CFAR), the system improves detection accuracy while maintaining low computational complexity. Unlike conventional fixed-threshold methods, the adaptive CFAR approach

dynamically adjusts the threshold based on noise conditions, reducing false alarms and missed detections. This makes the system more reliable and suitable for real-time wireless communication environments.

Furthermore, the effective utilization of TV white spaces enables the proposed system to provide cost-effective broadband connectivity in rural and underserved regions. The ability to dynamically access underutilized spectrum improves overall spectrum efficiency and supports the coexistence of Primary and Secondary Users without interference. This contributes to bridging the digital divide and enhancing communication infrastructure in remote areas. Overall, the work supports the development of next-generation wireless networks by promoting intelligent and adaptive spectrum management techniques.

1.5 Organization of the Report

The report is organized into four chapters to ensure a clear and structured presentation of the work. Chapter 2 provides a comprehensive review of existing literature related to Cognitive Radio and various spectrum sensing techniques. Chapter 3 presents the system design and implementation, including the proposed methodology and underlying algorithms. Chapter 4 discusses the simulation results, detailed performance analysis, and key findings, followed by conclusions and the future scope for further research.

CHAPTER 2: LITERATURE REVIEW

CHAPTER 2: LITERATURE REVIEW

2.1 Introduction

Cognitive Radio (CR) is an intelligent wireless communication paradigm that enables efficient utilization of the radio frequency spectrum by allowing unlicensed users to opportunistically access unused licensed bands. The concept of Cognitive Radio was first introduced by Joseph Mitola in 1999, marking a significant advancement in the evolution of wireless communication systems. It introduced the idea of radios that can sense, learn, and adapt to their surrounding environment, thereby improving spectrum efficiency. Since its inception, extensive research has been carried out in the areas of dynamic spectrum access, spectrum sensing, and adaptive communication techniques to address the growing problem of spectrum scarcity.

With the rapid expansion of wireless communication services such as mobile networks, Internet of Things (IoT), wireless sensor networks, and high-speed broadband applications, the demand for spectrum has increased significantly. However, traditional fixed spectrum allocation policies have resulted in inefficient utilization, where certain frequency bands remain underutilized across time and geographical regions, while others experience heavy congestion. This imbalance between spectrum demand and utilization has created the need for more flexible, adaptive, and intelligent spectrum management approaches.

Cognitive Radio addresses this issue by enabling real-time spectrum awareness, environment sensing, and adaptive transmission strategies. It allows Secondary Users (SUs) to detect unused spectrum bands, commonly referred to as spectrum holes or white spaces, and utilize them without causing interference to licensed Primary Users (PUs). This is achieved through key functionalities such as spectrum sensing, spectrum decision, spectrum sharing, and spectrum mobility. Among these, spectrum sensing plays a crucial role, as it determines the availability of spectrum and ensures reliable coexistence between users.

This capability is particularly important for emerging applications such as Wireless Regional Area Networks (WRAN), which aim to provide broadband connectivity in rural and underserved regions. By operating in underutilized TV white space frequency bands, WRAN systems can deliver wide-area coverage with minimal infrastructure. The efficient use of these bands, combined with intelligent spectrum management, makes Cognitive Radio a promising solution for bridging the digital divide and supporting next-generation wireless communication systems.

This chapter presents a detailed review of the fundamental concepts of Cognitive Radio, the IEEE 802.22 WRAN standard, and various spectrum sensing techniques. It also discusses their advantages, limitations, and practical considerations, which form the foundation for the proposed system design and implementation.

2.2 Cognitive Radio and Dynamic Spectrum Access

Cognitive Radio systems are designed to sense the surrounding radio environment, detect unused spectrum bands (commonly referred to as spectrum holes or white spaces), and dynamically adjust transmission parameters to utilize these bands without causing interference to licensed users. This capability is achieved through Dynamic Spectrum Access (DSA), which allows Secondary Users (SUs) to access licensed spectrum opportunistically when Primary Users (PUs) are inactive. By enabling such intelligent spectrum sharing, Cognitive Radio significantly improves overall spectrum efficiency and reduces the problem of spectrum scarcity.

The key functionalities of a Cognitive Radio system include spectrum sensing, spectrum decision, spectrum sharing, and spectrum mobility. Spectrum sensing is responsible for detecting the presence or absence of Primary Users in a given frequency band. Spectrum decision involves selecting the most suitable frequency band based on factors such as channel conditions and interference levels. Spectrum sharing ensures fair and efficient allocation of available spectrum among multiple Secondary Users, while spectrum mobility allows seamless switching to alternative frequency bands when the Primary User becomes active.

Among these functionalities, spectrum sensing plays a crucial role, as it determines the availability of spectrum and ensures that SU transmissions do not interfere with PU operations. Accurate sensing becomes challenging in practical scenarios due to noise uncertainty, fading, and hidden node problems. Therefore, advanced sensing techniques and adaptive algorithms are required to enhance detection performance under varying conditions.

Effective spectrum sensing techniques are essential for the successful deployment of Cognitive Radio systems. These techniques must achieve a balance between detection accuracy, computational complexity, and real-time performance. By combining efficient sensing methods with dynamic access strategies, Cognitive Radio systems can provide reliable communication while maximizing spectrum utilization in modern wireless networks.

2.3 IEEE 802.22 WRAN Standard

The IEEE 802.22 Wireless Regional Area Network (WRAN) standard is the first international standard specifically designed to incorporate Cognitive Radio technology. It operates in the TV broadcast frequency bands ranging from 54 MHz to 862 MHz, commonly known as TV white spaces. These frequency bands offer excellent propagation characteristics, such as low path loss and strong signal penetration through obstacles, making them highly suitable for providing broadband connectivity in rural and remote areas where conventional infrastructure is limited.

The IEEE 802.22 standard is designed to support long-range wireless communication, with coverage distances extending up to several tens of kilometers. It follows a point-to-multipoint architecture, where a base station communicates with multiple customer premises equipment (CPE). This architecture enables efficient distribution of broadband services over large geographical areas with minimal infrastructure requirements, making it a cost-effective solution for rural deployments.

The IEEE 802.22 standard emphasizes two critical aspects: incumbent protection and self-coexistence. Incumbent protection ensures that licensed users, such as TV broadcasters and wireless microphone users, are not affected by secondary transmissions. This is achieved through reliable spectrum sensing techniques and the use of geolocation databases to identify available channels. Self-coexistence mechanisms enable multiple WRAN systems to operate efficiently in the same geographical region without causing mutual interference, ensuring fair and efficient spectrum usage.

In addition, the standard incorporates advanced features such as dynamic frequency selection, adaptive power control, and centralized spectrum management. These features allow the system to respond to changing environmental conditions and optimize performance in real time. The use of Cognitive Radio technology enables WRAN systems to continuously monitor spectrum availability and adjust their operation accordingly.

Overall, IEEE 802.22 provides a robust framework for utilizing underutilized spectrum resources while ensuring protection of licensed users. Its ability to deliver long-range, cost-effective broadband connectivity makes it a promising solution for bridging the digital divide and supporting communication needs in underserved and remote regions.

2.4 Spectrum Sensing Techniques

Spectrum sensing is a fundamental functionality of Cognitive Radio systems, responsible for detecting the presence or absence of Primary Users (PUs) in a given frequency band. Accurate sensing ensures that Secondary Users (SUs) can opportunistically access unused spectrum without causing interference to licensed users. Over the years, several spectrum sensing techniques have been developed, each with its own advantages, limitations, and application scenarios. Figure 2.1 shows the different types of spectrum sensing techniques.

These techniques can be broadly classified into cooperative and non-cooperative approaches. Cooperative sensing improves detection accuracy by combining observations from multiple users, thereby reducing the effects of fading and shadowing. In contrast, non-cooperative sensing relies on individual user observations and is simpler to implement, but may suffer from reduced reliability in challenging environments. The choice of sensing technique depends on system requirements, trade-offs between complexity and accuracy, and the operating conditions.

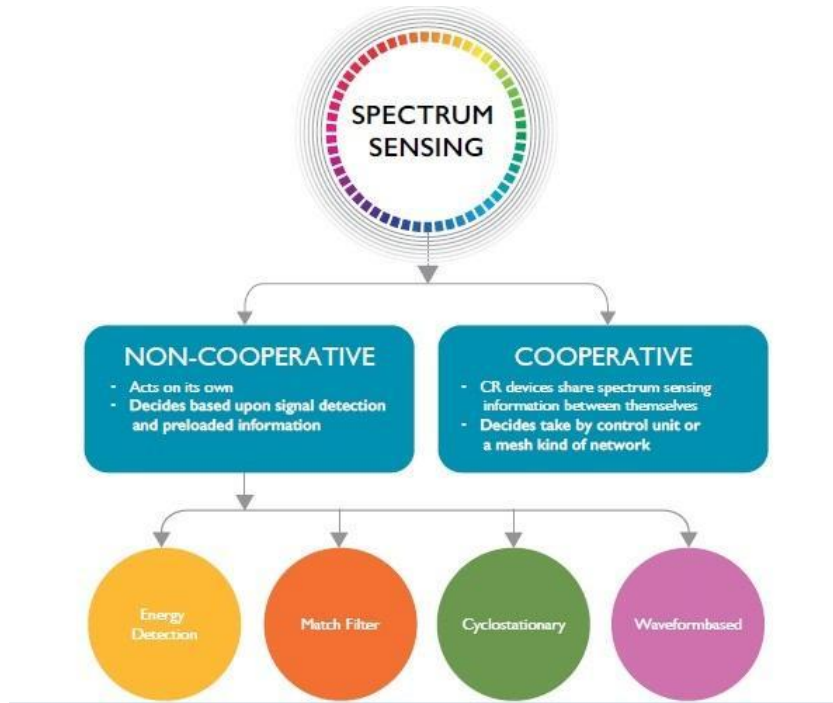


Figure 2.1: Spectrum Sensing Techniques

2.4.1 Energy Detection

Energy Detection is the most widely used spectrum sensing technique due to its simplicity, low computational complexity, and ease of implementation. In this method, the received signal is analyzed by computing its energy over a specific observation interval and comparing it with a predefined threshold. If the measured energy exceeds the threshold, the presence of a Primary User (PU) is assumed; otherwise, the band is considered free for Secondary User (SU) transmission.

The fundamental principle of energy detection is based on hypothesis testing, where two conditions are considered: the absence of the PU signal (noise only) and the presence of both signal and noise. The decision is made by evaluating the energy of the received signal samples. Since this technique does not require prior knowledge of the PU signal, such as modulation type, bandwidth, or structure, it is highly flexible and suitable for practical Cognitive Radio implementations.

One of the key advantages of energy detection is its applicability in real-time systems due to its simple mathematical operations and low processing requirements. It can be easily implemented for wideband sensing, where multiple frequency bands are monitored simultaneously. However, despite its advantages, the performance of energy detection is significantly affected by noise uncertainty, channel fading, and low Signal-to-Noise Ratio (SNR) conditions.

In practical scenarios, noise power is not constant and may vary over time, making it difficult to select an optimal fixed threshold. If the threshold is set too high, the system may fail to detect the presence of the PU (missed detection). Conversely, if the threshold is too low, the system may falsely detect the presence of a PU

even when the band is free (false alarm). These limitations reduce the reliability of conventional energy detection methods.

To overcome these challenges, adaptive thresholding techniques such as Constant False Alarm Rate (CFAR) are employed. CFAR dynamically adjusts the threshold based on real-time noise statistics, ensuring that the probability of false alarm remains approximately constant. This improves detection accuracy and makes the system more robust under varying environmental conditions. Overall, energy detection combined with adaptive thresholding provides an effective and practical solution for spectrum sensing in Cognitive Radio systems.

2.4.2 Matched Filter Detection

Matched Filter Detection is an optimal spectrum sensing technique when complete knowledge of the Primary User (PU) signal is available. It maximizes the Signal-to-Noise Ratio (SNR) by correlating the received signal with a known reference signal template. This process enhances the detectability of the signal even in the presence of noise, making it one of the most accurate sensing techniques. As a result, it provides fast detection with high reliability and is often considered the benchmark for performance comparison in spectrum sensing.

The working principle of matched filter detection is based on maximizing the output SNR by aligning the received signal with a predefined template that matches the PU signal characteristics. When the received signal matches the reference template, the output of the matched filter produces a peak, indicating the presence of the PU. This allows precise detection even at relatively low SNR conditions compared to simpler methods like energy detection.

One of the key advantages of this technique is its high detection accuracy and reduced sensing time. Since it directly exploits prior knowledge of the signal, it can achieve reliable detection with fewer samples. This makes it suitable for applications where quick and accurate detection is critical.

Despite its advantages, matched filter detection has several practical limitations. It requires exact knowledge of the PU signal, including modulation type, carrier frequency, pulse shape, and synchronization information. In real-world scenarios, such detailed information is often unavailable or difficult to obtain, limiting its applicability. Additionally, different signal types require separate matched filters, which increases hardware complexity and implementation cost.

Furthermore, the need for precise synchronization between the received signal and the reference template adds additional complexity to the system. Any mismatch in timing or frequency can degrade detection performance.

The requirement for multiple filters and synchronization mechanisms also increases power consumption, making it less suitable for low-power or resource-constrained devices.

Overall, while matched filter detection offers excellent performance in terms of accuracy and speed, its dependency on prior signal knowledge and high implementation complexity make it less practical for general-purpose Cognitive Radio systems. It is mainly suitable for scenarios where PU signal characteristics are known in advance and can be reliably utilized for detection.

2.4.3 Cyclostationary Feature Detection

Cyclostationary Feature Detection exploits the inherent periodicity present in modulated signals, such as carrier frequency, symbol rate, and cyclic prefixes. Unlike noise, which is typically random and does not exhibit periodic properties, communication signals possess cyclostationary features that can be used to distinguish them from noise. This property makes cyclostationary detection highly effective in identifying signals even in noisy environments.

The technique works by analyzing the cyclic autocorrelation and spectral correlation of the received signal. These cyclic features are unique to modulated signals and remain unaffected by noise, allowing reliable detection even when the Signal-to-Noise Ratio (SNR) is very low. As a result, cyclostationary detection can accurately identify the presence of a Primary User (PU) under challenging conditions where other techniques, such as energy detection, may fail.

One of the major advantages of this method is its ability to differentiate between different types of signals based on their cyclostationary properties. This makes it useful not only for detection but also for signal classification. It can distinguish between noise and interference, as well as between different modulation schemes, which adds an extra level of intelligence to spectrum sensing.

However, despite its advantages, cyclostationary feature detection involves complex signal processing operations such as spectral correlation function (SCF) computation and Fourier analysis. These operations require significant computational resources and longer observation times, making the method less suitable for real-time applications. The high complexity also leads to increased power consumption, which is a limitation for battery-operated devices.

Additionally, the implementation of this technique requires precise estimation of cyclic frequencies and careful signal processing design, which further increases system complexity. Due to these challenges, cyclostationary

detection is often used in applications where high accuracy is required and computational resources are available.

Overall, cyclostationary feature detection provides superior performance in terms of detection accuracy, especially under low SNR conditions. However, its high computational complexity and implementation overhead limit its practical use in simple and low-cost Cognitive Radio systems.

2.4.4 Eigenvalue-Based Detection

Eigenvalue-based detection methods rely on the statistical properties of the received signal's covariance matrix. These techniques analyze the distribution of eigenvalues obtained from the covariance matrix to determine the presence or absence of a Primary User (PU) signal. In the absence of a PU, the received signal consists mainly of noise, and the eigenvalues tend to be nearly equal. However, when a signal is present, the covariance structure changes, leading to noticeable differences among the eigenvalues.

The core idea behind this method is that the presence of a signal introduces correlation in the received samples, which is reflected in the eigenvalue distribution. By examining the ratio or spread between the largest and smallest eigenvalues, the system can reliably detect whether a signal is present. This makes eigenvalue-based detection effective in distinguishing between noise-only and signal-plus-noise conditions.

One of the major advantages of this technique is that it does not require prior knowledge of the PU signal characteristics or noise power. This makes it highly flexible and suitable for practical Cognitive Radio applications where such information is often unavailable. Additionally, it is more robust to noise uncertainty compared to energy detection, as it relies on statistical properties rather than absolute energy levels.

Another important benefit is its ability to perform well under low Signal-to-Noise Ratio (SNR) conditions. Since it utilizes covariance information, it can detect signals even when they are weak and buried in noise. This makes it a reliable alternative to conventional sensing methods in challenging environments.

Furthermore, the performance of this method depends on the number of samples used for covariance estimation. A larger number of samples improves detection accuracy but increases sensing time and computational load. Therefore, a trade-off exists between accuracy and complexity.

Overall, eigenvalue-based detection provides a robust and reliable approach for spectrum sensing without requiring prior signal knowledge. However, its high computational complexity and processing requirements limit its applicability in low-cost and real-time Cognitive Radio systems.

2.4.5 Comparison and Justification of Selected Technique

Each spectrum sensing technique offers a trade-off between complexity, accuracy, and practical feasibility. Matched filter and cyclostationary detection techniques provide high accuracy but suffer from high computational complexity and require prior knowledge of the signal. Eigenvalue-based methods are robust but computationally intensive. Energy detection, on the other hand, offers a simple and low-complexity solution but is sensitive to noise uncertainty.

In this project, **Energy Detection with adaptive CFAR thresholding** is selected as the preferred technique. This choice is motivated by its simplicity, low computational requirements, and suitability for real-time implementation in WRAN systems. By incorporating an adaptive threshold based on noise statistics, the limitations of conventional energy detection are effectively mitigated, resulting in improved detection performance under varying noise conditions. Furthermore, the integration of multi-band sensing enhances spectrum utilization by enabling simultaneous analysis of multiple frequency bands

2.5 Energy Detection and Challenges

Despite its advantages, conventional energy detection using a fixed threshold suffers from several limitations, particularly in environments with varying noise levels. When noise power fluctuates, a fixed threshold may lead to increased false alarms or missed detections, thereby reducing the reliability of spectrum sensing.

To overcome this limitation, adaptive thresholding techniques such as Constant False Alarm Rate (CFAR) have been proposed. CFAR dynamically adjusts the detection threshold based on the statistical properties of noise, ensuring that the probability of false alarm remains approximately constant. This significantly improves detection performance in practical scenarios where noise conditions are not stationary.

In addition, multi-band spectrum sensing has been introduced to improve spectrum utilization by analyzing multiple frequency bands simultaneously. This approach enables the system to identify and utilize available spectrum more efficiently compared to single-band sensing techniques.

2.6 Spectrum Occupancy models

Several models have been proposed in the literature to represent the activity of Primary Users (PUs) in Cognitive Radio systems. Among these, the Markov model is one of the most widely used due to its simplicity and effectiveness. However, other models such as Poisson processes, semi-Markov models, and machine

learning-based models are also used depending on the application requirements. A comparison of these models is presented below.

The **Markov model** assumes that the future state of the system depends only on the current state, making it a memoryless process. It is simple to implement and computationally efficient, which makes it highly suitable for simulation-based studies. It effectively captures the random ON/OFF behavior of spectrum usage, but may not fully represent complex traffic patterns with long-term dependencies.

The **Poisson model** is commonly used to represent random arrival processes, such as call arrivals in communication systems. It assumes that events occur independently with a constant average rate. While it is mathematically simple, it is not well-suited for modeling bursty or correlated spectrum usage patterns, as it does not capture state transitions explicitly.

The **Semi-Markov model** extends the Markov model by allowing variable state holding times, rather than assuming exponential distributions. This makes it more accurate in representing real-world traffic patterns, where the duration of ON and OFF states may not follow simple distributions. However, this increased accuracy comes at the cost of higher complexity and computational overhead.

Machine Learning-based models, such as those using neural networks or reinforcement learning, can learn complex and non-linear patterns in spectrum usage from data. These models can provide highly accurate predictions and adaptive decision-making capabilities. However, they require large amounts of training data, higher computational resources, and are more difficult to implement compared to traditional models.

In comparison, the Markov model offers a good balance between simplicity, computational efficiency, and modeling capability. For simulation-based Cognitive Radio systems such as WRAN, where real-time implementation and low complexity are important, the Markov model is an appropriate and practical choice.

2.7 Conclusions

The literature review highlights the importance of efficient spectrum sensing techniques in Cognitive Radio systems and emphasizes the limitations of conventional fixed-threshold energy detection methods. Various sensing techniques such as matched filter detection, cyclostationary feature detection, and eigenvalue-based detection offer improved accuracy but often come with increased computational complexity and implementation challenges. This makes them less suitable for real-time and low-cost systems.

Advanced techniques such as CFAR-based adaptive thresholding and multi-band spectrum sensing provide a more practical solution by improving detection performance under dynamic and uncertain noise conditions. These approaches help maintain a consistent false alarm rate while enhancing the ability to detect Primary User activity accurately. The combination of adaptability and efficiency makes these methods more suitable for real-world wireless environments.

In addition, the study of spectrum occupancy models highlights the importance of realistic modeling of Primary User behavior. Among various models, the Markov model provides a good balance between simplicity and effectiveness, making it suitable for simulation-based analysis. Understanding these models is essential for designing reliable spectrum sensing systems that can operate effectively under varying conditions.

Based on these observations, the present work focuses on implementing a multi-band adaptive energy detection scheme using CFAR to enhance detection reliability and spectrum utilization in a WRAN environment. The proposed approach achieves a balance between performance, computational complexity, and practical feasibility. Overall, the insights gained from this literature review form a strong foundation for the system design and implementation presented in the subsequent chapters.

CHAPTER 3: SYSTEM DESIGN AND IMPLEMENTATION

CHAPTER 3: SYSTEM DESIGN AND IMPLEMENTATION

3.1 Introduction

The proposed system models a Cognitive Radio (CR)-enabled Secondary User (SU) operating in a multi-band wireless environment for efficient spectrum utilization. The system integrates spectrum sensing, adaptive thresholding, and intelligent decision-making to dynamically access unused frequency bands without causing interference to the Primary User (PU). The overall architecture of the system is illustrated in **Figure 3.1**, which shows the flow of signal processing from the PU through the sensing, analysis, and decision-making blocks to the SU transmission.

In the proposed system, the Primary User acts as a licensed transmitter whose activity must be detected and protected. The PU signal propagates through a wireless channel that introduces impairments such as path loss, Rayleigh fading, and additive white Gaussian noise (AWGN). The received signal at the sensing node is a combination of the transmitted PU signal and noise, which reflects realistic wireless channel conditions.

The receiver, also referred to as the sensing node, captures and samples the incoming signal for further processing. This signal is then passed through a Fast Fourier Transform (FFT) or spectrum analyzer, where it is converted from the time domain to the frequency domain. The spectrum is divided into multiple bands, enabling multi-band spectrum sensing. This allows the system to analyze different frequency bands independently and identify available spectrum opportunities more efficiently.

Following this, the energy detector computes the energy of the received signal in each band. The computed energy values are compared with an adaptive threshold generated by the Constant False Alarm Rate (CFAR) block. The CFAR mechanism dynamically updates the threshold based on noise statistics, ensuring reliable detection even under varying noise conditions. A feedback loop is incorporated in the system, where noise-only samples are used to continuously refine the threshold, improving detection accuracy over time.

The decision logic block determines whether each band is occupied or free by comparing the energy values with the threshold. Based on this decision, the band selection module identifies a suitable free band for transmission. Before allowing transmission, a distance-based interference check is performed to ensure that the Secondary User is sufficiently far from the Primary User, thereby avoiding interference.

Finally, the Secondary User transmits only when both conditions—availability of a free band and satisfaction of the interference constraint—are met. This integrated approach enables efficient spectrum utilization while maintaining coexistence with licensed users. The system primarily focuses on adaptive energy detection using a Constant False Alarm Rate (CFAR) technique combined with multi-band spectrum sensing. This enables

reliable detection of spectrum holes under varying noise conditions and ensures practical applicability in Cognitive Radio-based WRAN systems

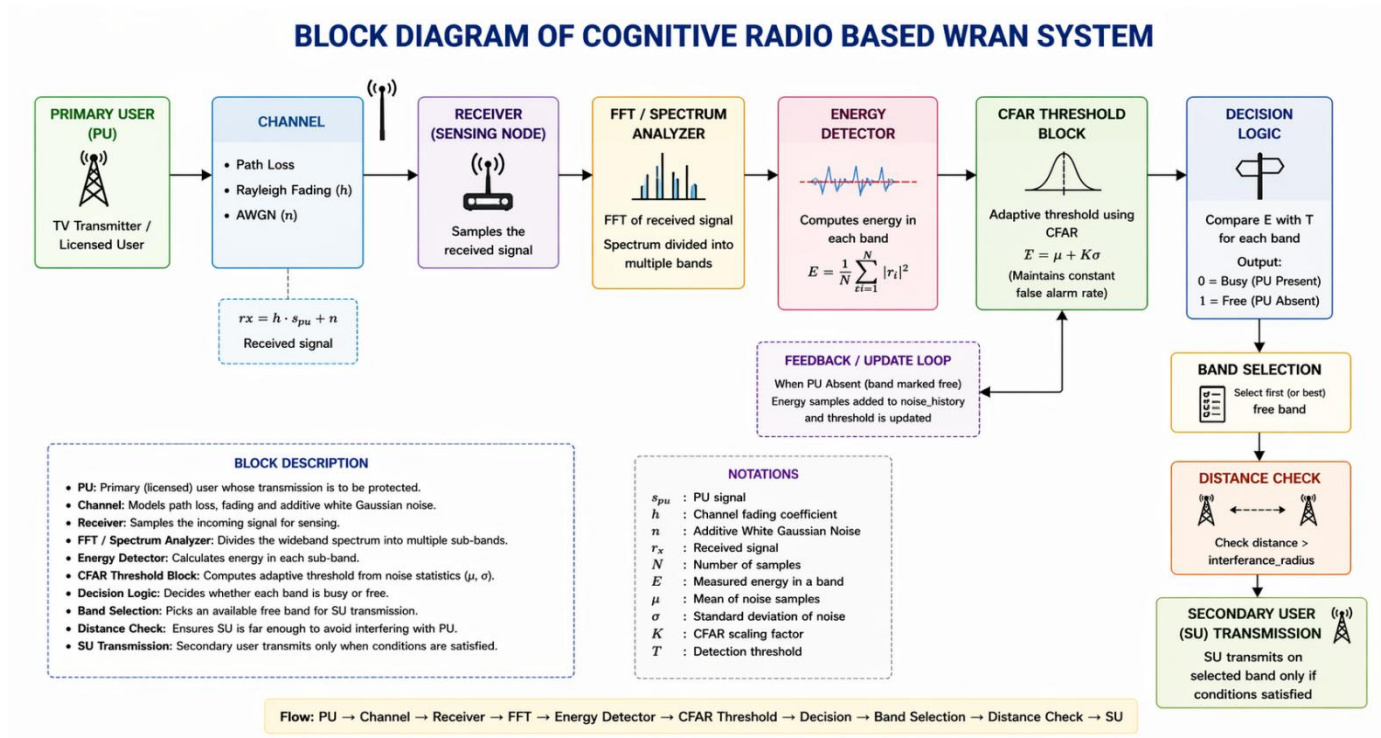


Figure 3.1: Block diagram of CR System

3.2 Primary User Activity Model

The activity of the Primary User (PU) is modeled using a two-state Markov chain representing the ON (active) and OFF (inactive) states. A Markov model is a stochastic process in which the future state depends only on the present state and not on the sequence of past states. This property makes it suitable for modeling time-varying systems such as wireless channel occupancy.

The transition probabilities are defined as $P_{01} = 0.1$, representing the probability of transition from the OFF state to the ON state, and $P_{10} = 0.3$, representing the probability of transition from the ON state to the OFF state. These probabilities control the behavior of the PU over time and simulate realistic spectrum usage patterns.

In this project, the Markov model is used to generate PU activity dynamically across multiple time slots, enabling the evaluation of the proposed spectrum sensing algorithm under varying conditions. The use of this model provides a more realistic representation compared to static or deterministic models, as it captures the random and time-dependent nature of spectrum occupancy. This helps in assessing the performance of the system in practical scenarios where PU activity is unpredictable.

3.3 Signal and Channel Model

The Primary User (PU) signal is modeled as a combination of two sinusoidal components of different frequencies to represent a practical communication signal. This representation captures the presence of multiple frequency components commonly found in real-world wireless transmissions. The generated signal provides a simple yet effective way to analyze the behavior of spectrum sensing techniques under realistic conditions.

The PU signal is transmitted through a wireless channel characterized by Rayleigh fading, which models multipath propagation effects. In practical wireless environments, signals travel through multiple paths due to reflections, scattering, and diffraction caused by obstacles such as buildings and terrain. As a result, the received signal experiences random fluctuations in amplitude and phase. Rayleigh fading is widely used to represent such non-line-of-sight (NLOS) conditions, making the simulation more realistic and challenging.

In addition to fading, additive white Gaussian noise (AWGN) with varying power is introduced to simulate environmental noise and interference. The noise power is allowed to vary over time to reflect real-world conditions where noise is not constant. This variation in noise significantly affects the signal quality and makes it difficult to distinguish between noise and actual PU signals, thereby increasing the complexity of the detection process.

The received signal at the sensing node is expressed as:

$$r(t) = h \cdot s_{PU}(t) + n(t)$$

where h represents the Rayleigh fading coefficient, $s_{PU}(t)$ is the transmitted PU signal, and $n(t)$ is the noise component. The fading coefficient h is a random variable that changes over time, introducing uncertainty in the received signal strength. The noise component $n(t)$ follows a Gaussian distribution and contributes to distortion in the observed signal.

The combined effects of fading and noise create a challenging environment for spectrum sensing. Under such conditions, simple detection methods may fail to accurately identify the presence of the PU. Therefore, adaptive techniques such as CFAR-based thresholding are required to dynamically adjust the detection threshold based on changing noise levels. This ensures more reliable detection and improves the robustness of the Cognitive Radio system in practical wireless scenarios. Additionally, the presence of channel variations may lead to sudden fluctuations in signal strength, making real-time adaptation essential for maintaining consistent sensing performance. Additionally, the presence of channel variations may lead to sudden fluctuations in signal strength, making real-time adaptation essential for maintaining consistent sensing performance.

3.4 Multi-Band Energy Detection

To enable efficient spectrum utilization, the received signal is analyzed using a multi-band approach. The signal is first transformed into the frequency domain using the Fast Fourier Transform (FFT), which converts the time-domain signal into its frequency components. This transformation allows the system to analyze different frequency bands independently and identify spectrum availability more effectively.

After the FFT operation, the entire frequency spectrum is divided into four equal frequency bands. Each band represents a specific portion of the spectrum, and the energy in each band is computed independently. This division enables simultaneous monitoring of multiple bands, reducing sensing time and improving the chances of detecting available spectrum opportunities compared to single-band sensing techniques.

The energy of the received signal is calculated as:

$$E = \frac{1}{N} \sum_{n=1}^N |r(n)|^2$$

where N is the number of samples and $r(n)$ represents the received signal. This energy calculation provides a measure of signal strength in each band and is used as the basis for detection.

Each band's energy is compared with a predefined threshold to determine whether the band is occupied or free. If the computed energy exceeds the threshold, the presence of the Primary User (PU) is assumed; otherwise, the band is considered available for Secondary User (SU) transmission. This decision-making process is simple and does not require prior knowledge of the PU signal, making it highly suitable for practical implementations.

Furthermore, the multi-band approach enhances spectrum efficiency by allowing the system to detect and utilize multiple spectrum holes simultaneously. It also improves system flexibility, as the Secondary User can select the most suitable band based on availability. However, the accuracy of this method depends on proper threshold selection, which is addressed using adaptive techniques such as CFAR in the proposed system. Additionally, analyzing multiple bands in parallel reduces sensing time and improves the overall responsiveness of the system to changing spectrum conditions. Overall, multi-band energy detection provides an effective and computationally efficient solution for dynamic spectrum sensing in Cognitive Radio environments.

3.5 Adaptive CFAR Threshold

To improve detection performance under varying noise conditions, an adaptive threshold based on the Constant False Alarm Rate (CFAR) technique is implemented. Unlike fixed-threshold methods, which assume constant noise levels, CFAR dynamically adjusts the threshold according to the statistical properties of the noise. The threshold is computed using the mean (μ) and standard deviation (σ) of recent noise-only energy samples as:

$$T = \mu + 2.5\sigma$$

This formulation ensures that the threshold adapts to fluctuations in noise power, making the detection process more reliable in practical environments where noise is unpredictable.

The system maintains a history of energy values corresponding to time slots where the Primary User (PU) is absent. These values are used to estimate the noise statistics required for threshold calculation. Once a sufficient number of samples (typically the last 20 values) are collected, the threshold is dynamically updated based on these statistics. During the initial stages, when insufficient data is available, a default threshold is used to allow the system to operate until enough noise samples are gathered.

The adaptive nature of CFAR enables the system to maintain a consistent false alarm rate even when noise conditions change. This prevents incorrect detection caused by sudden variations in noise power and improves the overall accuracy of spectrum sensing. By continuously updating the threshold, the system can effectively distinguish between noise and actual PU signals.

Additionally, the CFAR mechanism enhances system stability by reducing fluctuations in the threshold once sufficient data is collected. This leads to more consistent decision-making and minimizes detection errors over time. Overall, the use of adaptive CFAR thresholding significantly improves detection reliability, making it a suitable approach for dynamic and real-time Cognitive Radio environments.

3.6 Secondary User Decision and Band Selection

Based on the comparison between the measured energy and the adaptive threshold, each frequency band is classified as either occupied or free. If the energy in a band exceeds the threshold, it indicates the presence of a Primary User (PU), and the band is marked as occupied. Otherwise, the band is considered free and available for Secondary User (SU) transmission. This classification forms the basis for the decision-making process in the system.

If one or more bands are detected to be free, the system selects the first available band for transmission. This simple selection strategy ensures quick decision-making with minimal computational overhead. It also prioritizes lower-index bands, which are often more frequently available, thereby improving overall system efficiency. However, the system remains flexible, as it can select higher-index bands when lower bands are occupied.

Before allowing transmission, an interference constraint is applied by checking the distance between the Secondary User and the Primary User. A randomly generated distance is compared with a predefined interference radius (30 units). If the distance exceeds this threshold, the SU is permitted to transmit; otherwise, transmission is restricted to avoid interference. This mechanism ensures that the SU operates only when it is safe to do so, thereby protecting the PU from harmful interference.

The decision-making process is repeated for every time slot, allowing the system to dynamically adapt to changes in PU activity and spectrum availability. This continuous monitoring and decision update enable efficient utilization of available spectrum resources while maintaining reliable coexistence between users.

Additionally, the combination of spectrum sensing and interference-aware decision logic enhances the overall robustness of the system. It ensures that the SU not only selects free bands but also considers environmental constraints before transmission. Overall, this approach provides an effective balance between spectrum efficiency, interference avoidance, and system reliability in Cognitive Radio environments.

3.7 Algorithm Flowchart

The algorithmic flow of the proposed Cognitive Radio (CR) system is illustrated in Figure 3.2. The flowchart provides a clear and structured representation of the step-by-step operations involved in spectrum sensing, adaptive thresholding, and decision-making for Secondary User (SU) transmission. It helps in understanding how different components of the system interact to achieve efficient and interference-free spectrum utilization.

The process begins with system initialization, where key parameters such as the total number of time slots (N), initial time index ($t = 1$), and system variables are defined. After initialization, the algorithm enters a loop that executes for each time slot until the condition $t \leq N$ is satisfied. This loop-based execution enables the system to continuously monitor and adapt to dynamic spectrum conditions over time.

At each time slot, the Primary User (PU) activity is generated using a two-state Markov model. This model simulates realistic ON/OFF behavior of the PU by assigning probabilities for state transitions. Based on the

current state, the PU signal is either generated or set to zero. This step ensures that the system operates under realistic spectrum usage conditions.

The generated signal is then passed through a wireless channel model, where it is affected by Rayleigh fading and additive white Gaussian noise (AWGN). These channel effects introduce variations in signal amplitude and noise levels, making the detection process more challenging. The received signal at the sensing node is then processed for further analysis.

Next, energy detection is performed by computing the energy of the received signal. This energy value is used to determine the presence or absence of the PU. To improve detection accuracy under varying noise conditions, the CFAR-based adaptive threshold is updated using recent noise-only samples. If sufficient noise samples are available, the threshold is calculated using statistical measures; otherwise, a default threshold is used during the initial stages.

The signal is further processed using the Fast Fourier Transform (FFT) to convert it into the frequency domain. The spectrum is divided into multiple bands, and the energy of each band is computed independently. Each band's energy is compared with the adaptive threshold to determine whether it is occupied or free. This multi-band analysis enables efficient detection of spectrum holes across different frequency ranges.

If one or more bands are found to be free, the system selects the first available band for SU transmission. If no bands are free, the SU remains idle for that time slot. Before allowing transmission, an interference constraint is applied by checking the distance between the SU and PU. If the distance exceeds a predefined interference radius, transmission is permitted; otherwise, it is restricted to avoid interference.

Finally, the time index is incremented, and the process repeats for the next time slot. This iterative execution continues until all time slots are completed, after which the system outputs results such as energy values, threshold variations, band activity, ROC curves, and SU band selection.

The flowchart effectively illustrates the logical sequence and interaction between different modules of the system. It highlights how sensing, adaptation, and decision-making are integrated to achieve reliable detection and efficient spectrum utilization. Overall, the algorithm ensures dynamic operation, robustness under varying conditions, and safe coexistence between Primary and Secondary Users in Cognitive Radio environments.

FLOWCHART OF COGNITIVE RADIO (CR) ALGORITHM

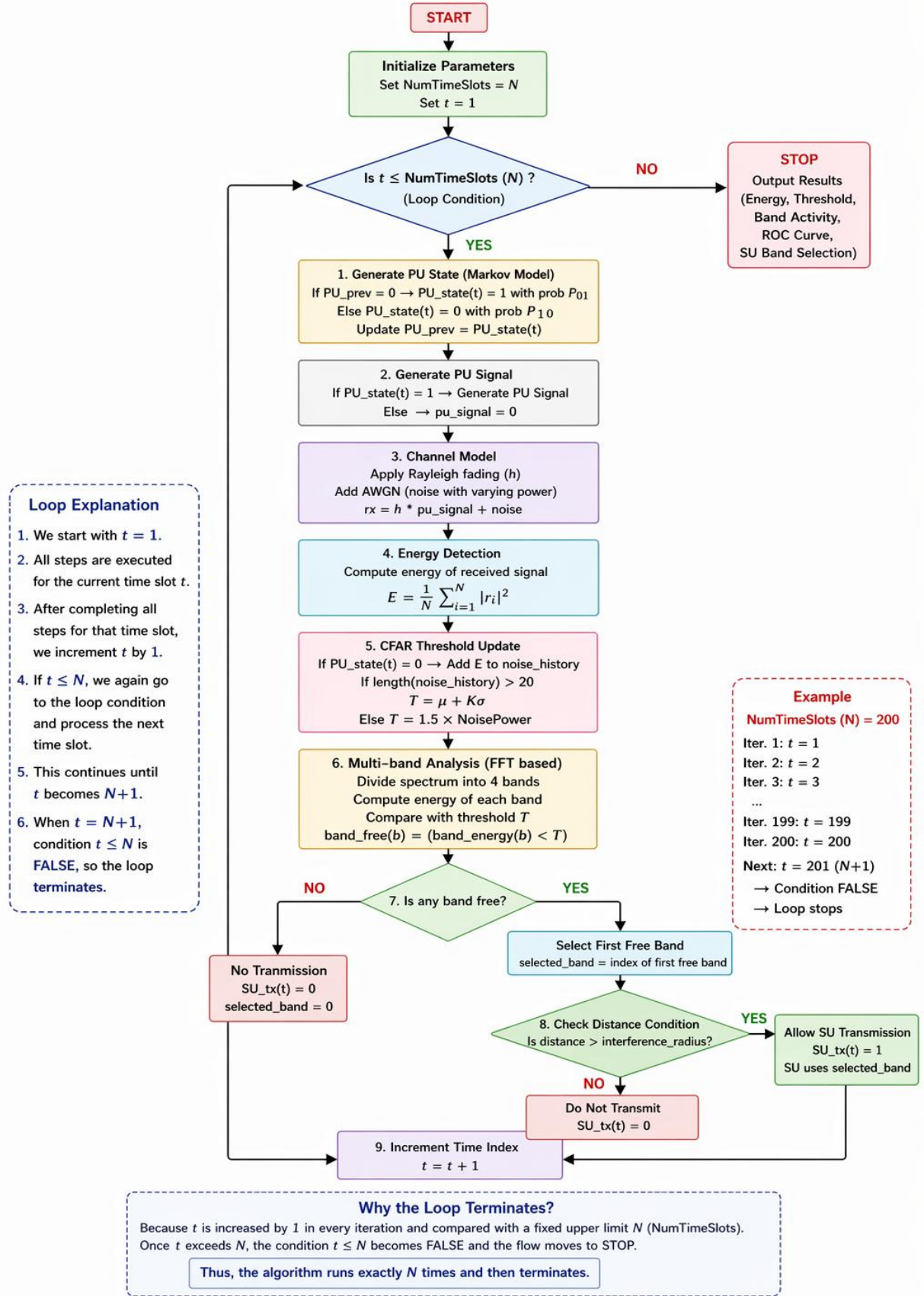


Figure 3.2: Algorithm Flowchart

3.8 MATLAB Implementation

The complete system is implemented using MATLAB. The main function `cr_final_subplots()` performs the simulation over multiple time slots and generates all relevant outputs, including energy variation, adaptive threshold behavior, band-wise activity of PU and SU, ROC curve, and selected bands for SU transmission.

The implementation incorporates key components such as the Markov PU model, Rayleigh fading channel, CFAR-based thresholding, FFT-based multi-band analysis, and interference-aware decision logic. The detailed MATLAB code is provided in **Appendix A**.

3.9 Conclusions

The designed system successfully demonstrates adaptive multi-band spectrum sensing for a Cognitive Radio-based WRAN environment. By integrating CFAR-based thresholding with multi-band energy detection and intelligent decision-making, the system achieves efficient spectrum utilization while ensuring minimal interference to the Primary User. The results validate the effectiveness of the proposed approach under varying channel and noise conditions.

CHAPTER 4: RESULTS AND CONCLUSIONS

CHAPTER 4: RESULTS AND CONCLUSIONS

4.1 Introduction

This chapter presents the simulation results obtained from the MATLAB-based implementation of the proposed Cognitive Radio (CR) system for Wireless Regional Area Network (WRAN) applications. The system is evaluated under dynamic operating conditions by modeling Primary User (PU) activity using a Markov process, incorporating channel effects such as Rayleigh fading, and introducing noise with varying power levels. These conditions closely resemble real-world wireless environments and help in assessing the robustness, reliability, and adaptability of the proposed approach under practical scenarios.

The performance of the system is analyzed using multi-band energy detection combined with an adaptive thresholding technique based on Constant False Alarm Rate (CFAR). The adaptive threshold dynamically adjusts according to noise variations, improving the accuracy of spectrum sensing compared to conventional fixed-threshold methods. This enables consistent detection performance even under challenging conditions such as low Signal-to-Noise Ratio (SNR) and fluctuating channel conditions.

In addition, the behavior of the Secondary User (SU) in selecting appropriate frequency bands without interfering with the PU is studied in detail. The system demonstrates intelligent decision-making by continuously identifying available spectrum holes and allocating them efficiently. The inclusion of an interference-aware mechanism ensures proper coexistence between Primary and Secondary Users, thereby maintaining reliable and interference-free communication.

Key performance metrics such as energy variation over time, adaptive threshold behavior, band-wise activity of PU and SU, and Receiver Operating Characteristic (ROC) analysis are presented and discussed comprehensively. The results also highlight the trade-off between detection probability and false alarm rate, which is crucial for evaluating sensing performance. Overall, the findings validate the effectiveness, stability, and practical applicability of the proposed approach for Cognitive Radio-based WRAN systems.

4.2 PU vs SU Activity per Band

Figure 4.1 shows the activity of the Primary User (PU) in red and the Secondary User (SU) in blue across four frequency bands over 200 time slots. The results clearly indicate that the SU successfully transmits only in the frequency bands where the PU is inactive, thereby avoiding interference with licensed users. This demonstrates the effectiveness of the proposed spectrum sensing and decision-making mechanism.

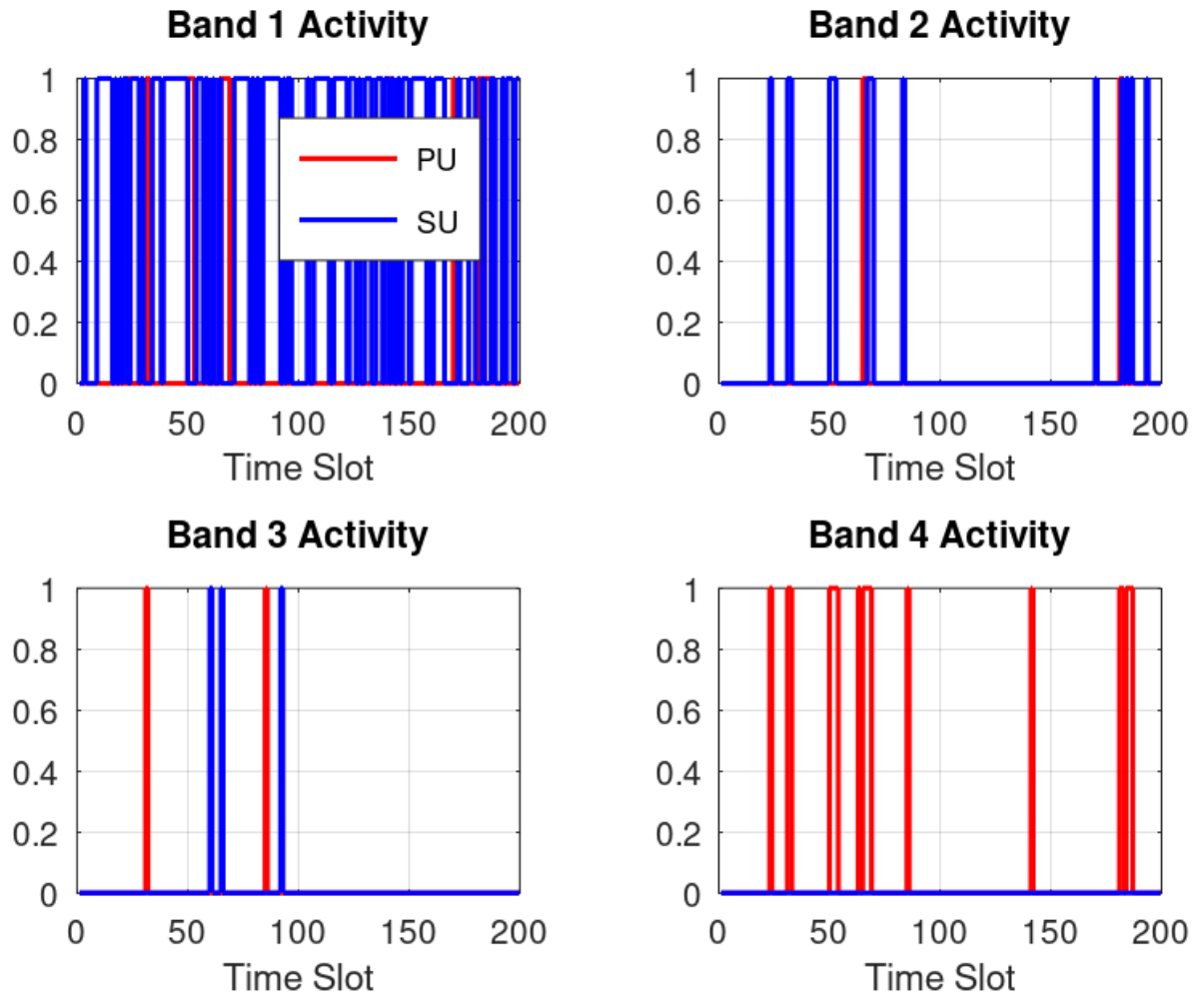


Figure 4.1: PU vs SU Activity per Band.

From the figure, it can be observed that Band 1 has frequent availability, resulting in higher SU activity. Band 2 exhibits moderate usage with occasional SU transmissions, while Band 3 shows limited activity due to fewer available spectrum opportunities. In contrast, Band 4 is predominantly occupied by the PU, which significantly restricts SU transmission in that band. This variation across bands highlights the dynamic nature of spectrum availability.

The results also show that the system adapts efficiently to changing PU activity over time. Whenever the PU becomes active, the SU immediately stops transmission in that band and switches to another available band if possible. This dynamic behavior ensures continuous communication while maintaining protection for the Primary User.

Furthermore, the clear separation between PU and SU activities in the graph indicates accurate spectrum sensing and reliable decision-making. The absence of overlapping transmissions confirms that the proposed system effectively avoids interference and utilizes spectrum holes efficiently. Overall, the results validate the

capability of the system to achieve intelligent spectrum access and optimal utilization of available frequency bands.

4.3 Energy vs Adaptive Threshold

Figure 4.2 illustrates how the adaptive threshold (red) closely follows the noise floor while the energy (blue) rises significantly when the Primary User (PU) is active. This graph shows the variation of the measured signal energy and the adaptive CFAR threshold over different time slots. Initially, the threshold is set to a higher constant value due to insufficient noise samples. As the simulation progresses and more noise-only samples are collected, the threshold becomes adaptive and stabilizes around a certain level based on noise statistics.

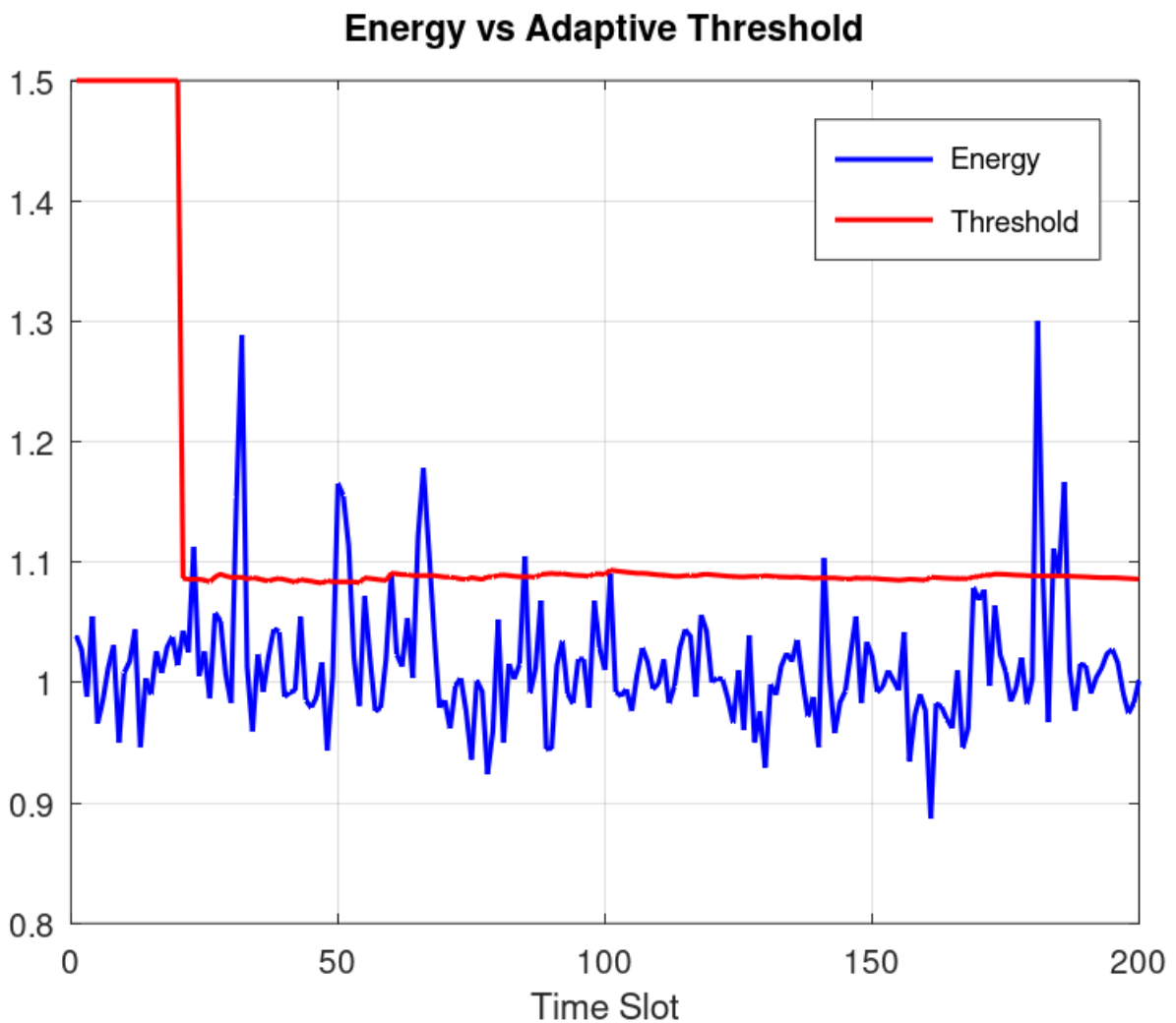


Figure 4.2: Energy vs Adaptive Threshold Plot.

The energy values fluctuate due to the combined effects of noise and channel conditions such as fading. When the PU is present, the energy rises above the threshold, indicating successful detection. When the PU is absent,

the energy remains below the threshold, allowing the system to identify spectrum holes accurately. This clear distinction between energy and threshold ensures reliable spectrum sensing.

The adaptive threshold effectively tracks variations in the noise floor, preventing incorrect decisions caused by noise uncertainty. Unlike fixed-threshold methods, which may result in high false alarm rates or missed detections, the CFAR-based approach dynamically adjusts to changing conditions. This improves the robustness and consistency of the sensing process.

Furthermore, the graph demonstrates that the system maintains a stable threshold after sufficient learning, reducing fluctuations and improving decision reliability. The occasional spikes in energy correspond to PU activity, which are correctly detected by the system. Overall, the results validate that adaptive CFAR thresholding enhances detection performance and enables efficient spectrum utilization in dynamic wireless environments.

4.4 ROC Curve Analysis

Figure 4.3 shows the Receiver Operating Characteristic (ROC) curve at SNR = -10 dB, indicating good detection performance with a reasonable false alarm rate. The ROC curve represents the trade-off between the Probability of Detection (P_d) and the Probability of False Alarm (P_{fa}). An efficient spectrum sensing system aims to achieve high detection probability while maintaining a low false alarm rate, ensuring optimal utilization of available spectrum.

From the graph, it is observed that the detection probability increases as the false alarm rate increases. Even at lower values of P_{fa} , the system maintains a relatively high P_d , which indicates strong detection capability under strict conditions. This behavior is desirable in Cognitive Radio systems, as it ensures that Primary Users are accurately detected while minimizing unnecessary restrictions on Secondary Users utilization.

The smooth and gradually increasing nature of the ROC curve reflects the stability and consistency of the sensing algorithm. As P_{fa} increases, the curve approaches higher values of P_d , indicating that near-perfect detection can be achieved when a slightly higher false alarm rate is acceptable. This demonstrates the inherent trade-off between sensitivity and reliability in detection systems

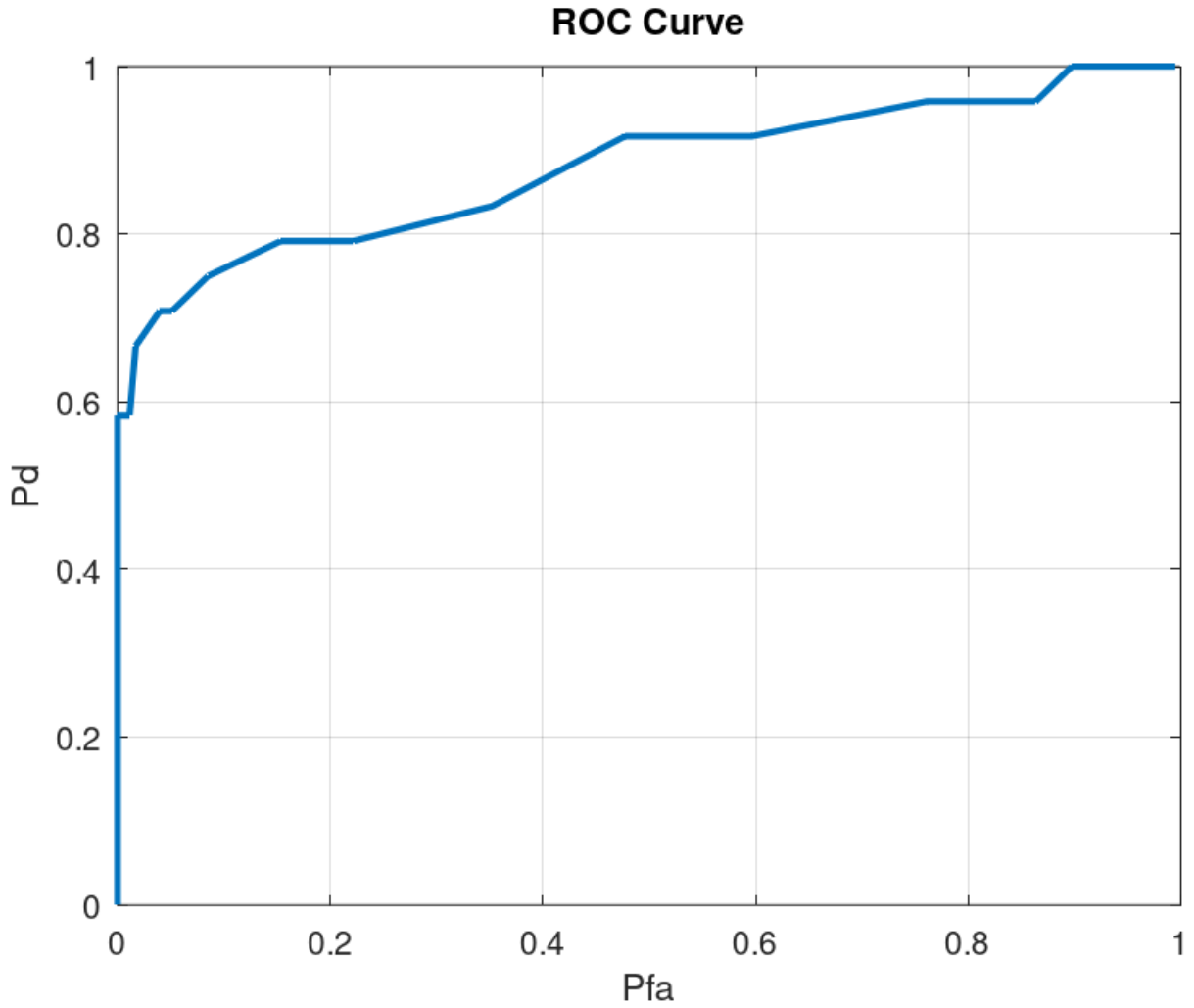


Figure 4.3: Receiver Operating Characteristic (ROC) Curve.

The performance at a low SNR of -10 dB highlights the robustness of the proposed approach under challenging noise conditions. The adaptive CFAR-based thresholding enhances the system's ability to distinguish between noise and signal effectively. Overall, the ROC analysis confirms that the proposed method achieves a balanced and reliable detection performance, making it suitable for practical Cognitive Radio-based WRAN applications.

4.5 Band Allocation by Secondary User

Figure 4.4 depicts how the Secondary User (SU) dynamically switches between different frequency bands (1–4) based on spectrum availability. This graph shows the band selected by the SU over different time slots, reflecting the adaptive nature of the proposed system. The SU continuously monitors all bands and selects an available band for transmission whenever the Primary User (PU) is inactive.

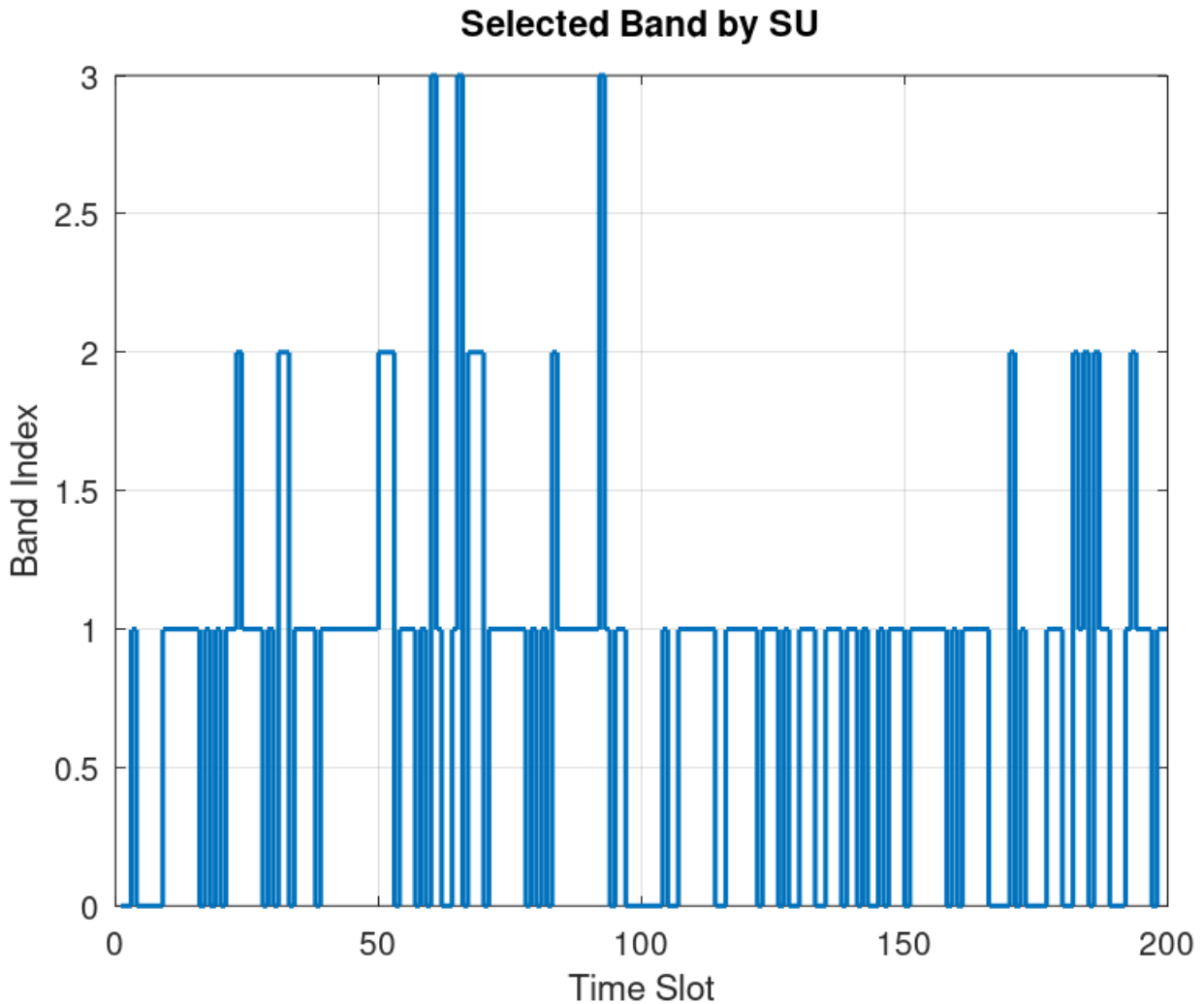


Figure 4.4: SU Band Selection

Most of the time, lower-index bands are selected because the algorithm is designed to choose the first available free band. This results in frequent usage of Band 1 when it is available. In some instances, higher-index bands such as Band 2 or Band 3 are selected when lower bands are occupied by the PU. This behavior indicates that the system efficiently utilizes all available spectrum resources rather than relying on a fixed band.

The dynamic switching pattern observed in the graph demonstrates the system's ability to quickly respond to changes in PU activity. Whenever a band becomes occupied, the SU immediately shifts to another free band, ensuring uninterrupted communication. This reduces idle time and improves overall spectrum utilization efficiency.

Additionally, the presence of gaps (where no band is selected) indicates time slots where no spectrum holes are available or interference conditions are not satisfied. This shows that the system prioritizes interference avoidance over transmission, thereby protecting Primary Users. Overall, the results confirm that the proposed approach enables intelligent band selection, efficient resource utilization, and reliable operation in a dynamic spectrum environment.

4.6 Performance Discussion

The performance of the proposed Cognitive Radio system is evaluated based on its ability to accurately detect Primary User (PU) activity, adapt to varying noise conditions, and efficiently utilize available spectrum bands. The use of an adaptive threshold based on the Constant False Alarm Rate (CFAR) technique significantly improves the robustness of the system against noise uncertainty. Unlike fixed-threshold methods, which are highly sensitive to noise fluctuations, the CFAR-based approach dynamically adjusts the threshold using real-time noise statistics, thereby maintaining a consistent false alarm rate and enhancing detection reliability across different operating conditions.

The multi-band spectrum sensing approach further improves system performance by enabling simultaneous analysis of multiple frequency bands. This allows the Secondary User (SU) to identify and exploit spectrum holes more effectively compared to single-band sensing techniques. As observed from the results, the SU dynamically selects available bands and avoids those occupied by the PU, leading to efficient spectrum utilization. This capability ensures that the system can maximize the use of available resources while minimizing idle spectrum.

Additionally, the incorporation of an interference-aware decision mechanism, based on a distance constraint, ensures that SU transmission does not interfere with the PU. This highlights the system's ability to maintain coexistence with licensed users while maximizing SU throughput. The system prioritizes interference avoidance, which is essential for practical deployment in real-world environments where protecting licensed users is critical.

The ROC analysis further demonstrates a favorable trade-off between detection probability and false alarm rate, confirming the effectiveness of the proposed adaptive sensing method. The system performs well even under low Signal-to-Noise Ratio (SNR) conditions, indicating its robustness in challenging environments. Moreover, the consistent performance observed across different time slots shows the stability and reliability of the proposed approach.

Overall, the system achieves reliable detection, efficient band allocation, and interference-free communication. The combination of adaptive thresholding, multi-band sensing, and intelligent decision-making makes the

proposed method suitable for practical Cognitive Radio-based WRAN applications, offering a balanced solution in terms of performance, complexity, and scalability.

4.7 Conclusions and Future Scope

The proposed project successfully demonstrates an efficient Cognitive Radio-based spectrum sensing and access mechanism suitable for Wireless Regional Area Network (WRAN) applications. By integrating multi-band energy detection with an adaptive CFAR-based thresholding technique, the system effectively identifies spectrum opportunities under varying noise conditions. The use of a Markov model for Primary User activity, along with realistic channel modeling including fading and noise variations, ensures that the system performance closely reflects practical wireless environments.

The simulation results confirm that the proposed approach achieves reliable detection of Primary User activity, reduces false alarms, and enables dynamic spectrum access for the Secondary User without causing harmful interference. The system also demonstrates the ability to adapt to changing spectrum conditions in real time and efficiently allocate available frequency bands. The multi-band sensing approach significantly improves spectrum utilization, while the adaptive thresholding mechanism enhances detection accuracy under uncertain noise conditions. Overall, the proposed method provides a balanced solution in terms of detection performance, computational efficiency, and practical feasibility.

In addition, the results highlight the stability and consistency of the system across different time slots and varying channel conditions. The integration of intelligent decision-making mechanisms ensures that the Secondary User utilizes only available spectrum opportunities while maintaining coexistence with licensed users. This makes the system highly suitable for deployment in dynamic wireless environments, particularly in rural and underserved regions where efficient spectrum utilization is essential.

Future Scope

Although the proposed system demonstrates effective performance in a simulation environment, several enhancements can be explored in future work to further improve its capabilities. The implementation of the system on Software Defined Radio (SDR) platforms such as USRP can enable real-time validation and facilitate practical deployment in real-world scenarios. This would allow evaluation under actual channel conditions and user mobility.

Advanced spectrum sensing techniques, such as cyclostationary feature detection, eigenvalue-based detection, or hybrid sensing methods, can be incorporated to improve detection accuracy, particularly under low Signal-

to-Noise Ratio (SNR) conditions. Furthermore, machine learning and artificial intelligence-based approaches can be integrated to enable intelligent and adaptive decision-making based on environmental patterns and historical data.

Future work can also focus on cooperative spectrum sensing, where multiple Cognitive Radio users share sensing information to improve detection reliability and reduce the impact of fading and shadowing. Additionally, optimization of band selection strategies can be explored to enhance throughput and fairness among multiple Secondary Users.

REFERENCES

- [1] J. Mitola and G. Q. Maguire, “Cognitive radio: making software radios more personal,” *IEEE Personal Communications*, vol. 6, no. 4, pp. 13–18, Aug. 1999.
- [2] IEEE Std 802.22-2011, “IEEE Standard for Information Technology—Local and metropolitan area networks—Specific requirements—Part 22: Cognitive Wireless RAN Medium Access Control (MAC) and Physical Layer (PHY) specifications: Policies and Procedures for Operation in the TV Bands,” 2011.
- [3] S. Haykin, “Cognitive radio: brain-empowered wireless communications,” *IEEE Journal on Selected Areas in Communications*, vol. 23, no. 2, pp. 201–220, Feb. 2005.
- [4] T. Yucek and H. Arslan, “A survey of spectrum sensing algorithms for cognitive radio applications,” *IEEE Communications Surveys & Tutorials*, vol. 11, no. 1, pp. 116–130, First Quarter 2009.
- [5] MATLAB Documentation, Signal Processing Toolbox.

APPENDIX A – MATLAB CODE

Appendix A

MATLAB Code for Cognitive Radio Spectrum Sensing Simulation

```
matlab
function cr_final_subplots()
% CR_FINAL_SUBPLOTS: Simulates a Cognitive Radio (CR) environment.
% It models a Primary User (PU) with Markovian behavior, an Adaptive
% Energy Detector (Sensing), and a Secondary User (SU) that makes
% decisions based on spectrum availability and distance.

clc; clear; close all;

%% ===== 1. SYSTEM PARAMETERS =====
N = 1024;          % Number of samples per time slot
NumTimeSlots = 200; % Total duration of the simulation
Fs = 1e6;          % Sampling frequency (1 MHz)

% Power and SNR Settings
NoisePower = 1;    % Reference noise floor
SNR_dB = -10;      % Signal-to-Noise Ratio in dB (Low SNR for testing robustness)
SignalPower = 10^(SNR_dB/10)*NoisePower;

% Markov Chain Transition Probabilities (Temporal behavior of PU)
P01 = 0.1;         % Probability Idle -> Busy (Birth rate)
P10 = 0.3;         % Probability Busy -> Idle (Death rate)

% Environment Factors
rho = 1.02;        % Noise uncertainty factor
numBands = 4;      % Number of frequency sub-bands to monitor
interference_radius = 30; % Physical distance threshold for SU transmission

%% ===== 2. MEMORY ALLOCATION =====
PU_state = zeros(1,NumTimeSlots); % Ground truth of PU (1=ON, 0=OFF)
```

```

SU_tx = zeros(1,NumTimeSlots);      % Logic for SU transmission
energy_vec = zeros(1,NumTimeSlots);  % Total energy calculated per slot
threshold_vec = zeros(1,NumTimeSlots); % Calculated adaptive threshold

PU_band = zeros(numBands, NumTimeSlots); % Per-band PU activity
SU_band_matrix = zeros(numBands, NumTimeSlots); % Per-band SU allocation

noise_history = []; % Buffer to store energy samples when PU is absent
PU_prev = 0;      % Initial Markov state

%% ===== 3. MAIN SIMULATION LOOP =====
for t = 1:NumTimeSlots

    %% --- A. MARKOV PU STATE GENERATION ---
    % Simulates the "bursty" nature of a licensed user
    if PU_prev == 0
        PU_state(t) = rand < P01;
    else
        PU_state(t) = ~(rand < P10);
    end
    PU_prev = PU_state(t);

    %% --- B. SIGNAL GENERATION ---
    time = (0:N-1)/Fs;
    if PU_state(t)==1
        % Generate a PU signal with two distinct frequency components
        pu_base = cos(2*pi*20e3*time) + cos(2*pi*80e3*time);
        % Normalize to desired SignalPower
        pu_signal = sqrt(SignalPower/mean(pu_base.^2))*pu_base;
    else
        pu_signal = zeros(1,N); % No PU signal
    end

    %% --- C. CHANNEL MODEL (Fading + Noise) ---
    h = (randn + 1j*randn)/sqrt(2); % Rayleigh Fading Channel
    % Simulate noise uncertainty (fluctuating noise floor)

```



```

noise_pow = NoisePower*(1 + (rho-1)*(2*rand-1));
noise = sqrt(noise_pow/2)*(randn(1,N)+1j*randn(1,N));

% Signal at the receiver (Sensing observation)
rx = h*pu_signal + noise;

%% --- D. ENERGY DETECTION ---
% Calculate Mean Square of the received samples
energy = mean(abs(rx).^2);
energy_vec(t) = energy;

%% --- E. ADAPTIVE CFAR THRESHOLDING ---
% If we assume PU is absent (H0), we update our noise statistics
if PU_state(t)==0
    noise_history = [noise_history energy];
end

% Threshold updates once we have enough data (sliding window concept)
if length(noise_history)>20
    mu = mean(noise_history);
    sigma = std(noise_history);
    % Threshold set to maintain a low False Alarm rate (mu + k*sigma)
    Threshold = mu + 2.5*sigma;
else
    % Default fallback threshold
    Threshold = 1.5*NoisePower;
end
threshold_vec(t) = Threshold;

%% --- F. MULTI-BAND SPECTRUM ANALYSIS (FFT) ---
Y = fft(rx);
PSD = abs(Y).^2/N; % Power Spectral Density estimation

band_size = floor(N/numBands);
band_energy = zeros(1,numBands);

```

```

for b=1:numBands
    idx1=(b-1)*band_size+1;
    idx2=b*band_size;
    % Measure energy in specific frequency bins
    band_energy(b)=mean(PSD(idx1:idx2));
end

% Determine if a band is free based on the local threshold
band_free = band_energy < Threshold;

% Track PU ground truth per band for visualization
for b = 1:numBands
    if PU_state(t)==1 && band_energy(b) > Threshold
        PU_band(b,t) = 1;
    else
        PU_band(b,t) = 0;
    end
end

%% --- G. SU DECISION & RESOURCE ALLOCATION ---
if any(band_free)
    selected_band = find(band_free,1); % Select the first available hole
    detected_free = 1;
else
    detected_free = 0;
    selected_band = 0;
end

% Check spatial condition (SU must be far enough to avoid interference)
distance = rand*100;

if detected_free==1 && distance > interference_radius
    SU_tx(t)=1;
    SU_band_matrix(selected_band,t)=1;
else
    SU_tx(t)=0;

```

end

end

C

%% Figure 1: Activity Per Band

figure('Name', 'Multi-Band Activity Tracking');

for b = 1:numBands

 subplot(2,2,b);

 stairs(PU_band(b,:), 'r', 'LineWidth', 1.5); hold on;

 stairs(SU_band_matrix(b,:), 'b', 'LineWidth', 1.5);

 ylim([-0.2 1.2]);

 grid on;

 title(['Band ', num2str(b), ' Activity']);

 if b == 1, legend('PU (Licensed)', 'SU (Secondary)'); end

end

%% Figure 2: Energy vs Threshold

figure('Name', 'Spectrum Sensing Performance');

plot(energy_vec, 'b', 'LineWidth', 1.5); hold on;

plot(threshold_vec, 'r', 'LineWidth', 1.5);

legend('Observed Energy', 'Adaptive Threshold');

title('Energy Detection with Adaptive CFAR Threshold');

xlabel('Time Slot'); ylabel('Energy Level'); grid on;

%% Figure 3: ROC Curve (Detection Reliability)

thresholds = linspace(min(energy_vec), max(energy_vec), 40);

Pd = zeros(size(thresholds));

Pfa = zeros(size(thresholds));

for i=1:length(thresholds)

 th = thresholds(i);

 decision = energy_vec > th;

 Pd(i) = sum(decision(PU_state==1)==1)/sum(PU_state==1);

 Pfa(i) = sum(decision(PU_state==0)==1)/sum(PU_state==0);

end

```
figure('Name', 'ROC Curve');  
plot(Pfa,Pd,'m','LineWidth',2);  
xlabel('Probability of False Alarm (Pfa)');  
ylabel('Probability of Detection (Pd)');  
title('Receiver Operating Characteristic (ROC)');  
grid on;
```

%% Figure 4: SU Band Selection Over Time

```
figure('Name', 'Resource Allocation');  
stairs(sum(SU_band_matrix.*(1:numBands)',1),'Color',[0 0.5 0],'LineWidth',1.5);  
xlabel('Time Slot'); ylabel('Band Index (1-4)');  
title('Band Index Selected by SU for Opportunistic Access');  
ylim([0 numBands+1]); grid on;  
end
```